Power & Risks of AI

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What is Artificial Intelligence (AI)?

...perform tasks normally requiring human **reasoning**, **learning**, and **problem solving**

https://www.media.mit.edu/tools/ai-glossary-dictionary

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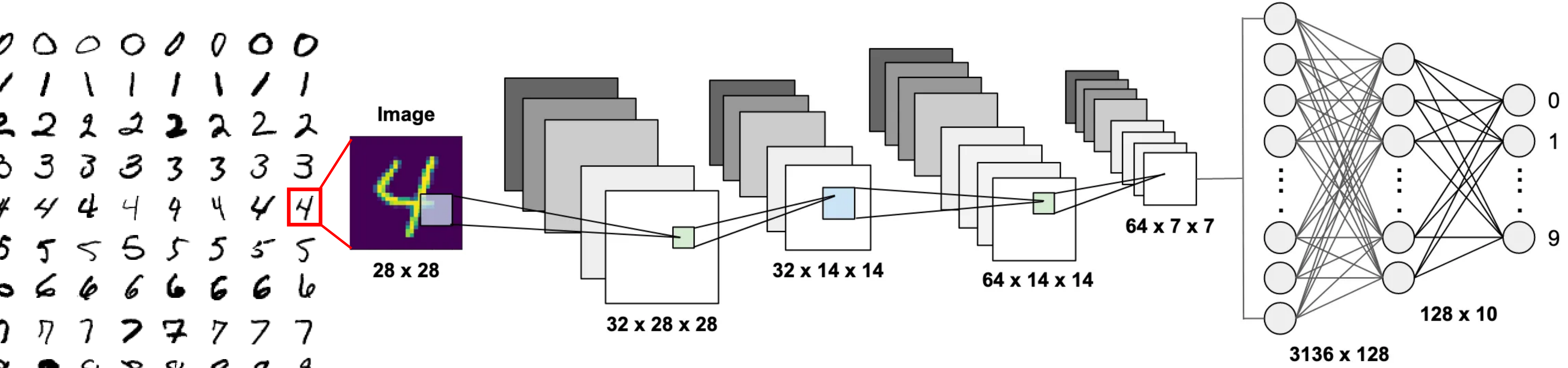
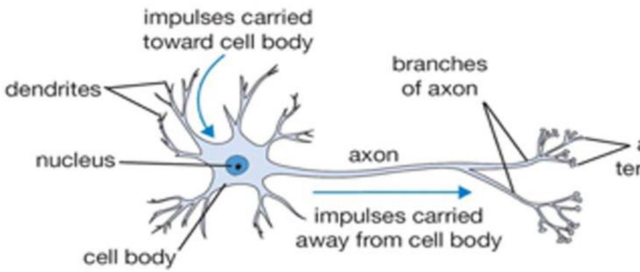
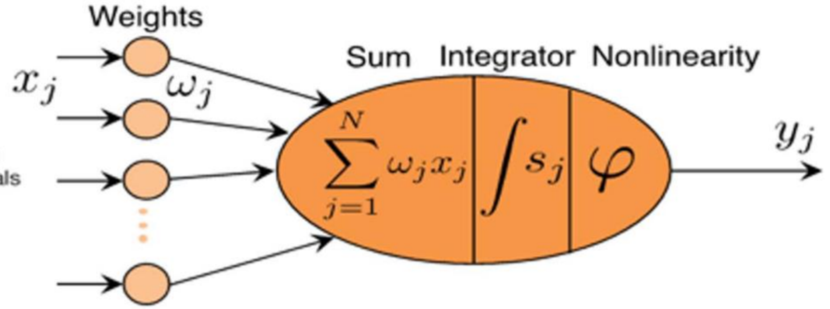
Artificial Neural Networks (ANNs)







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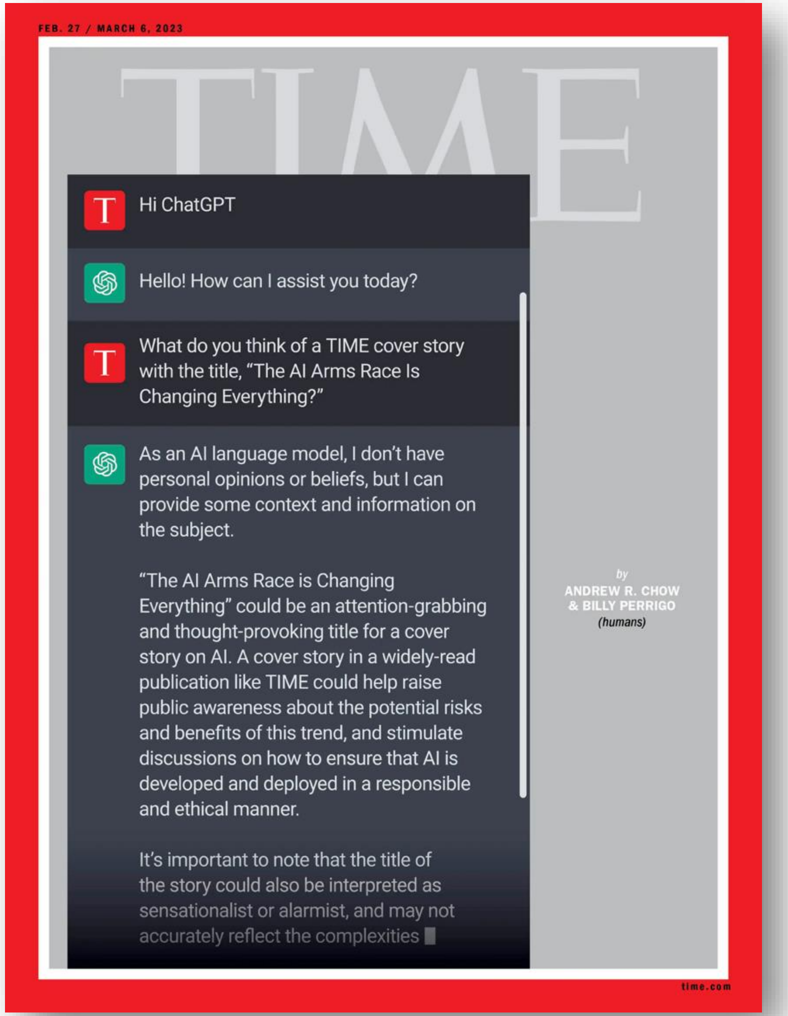
Picture from: L. Xu *et al.*, "Scalable Networks of Neuromorphic Photonic Integrated Circuits," in *IEEE Journal of Selected Topics in Quantum Electronics*

What caused the recent AI boom?

- Advancements in hardware
- Data explosion
- Algorithmic improvements



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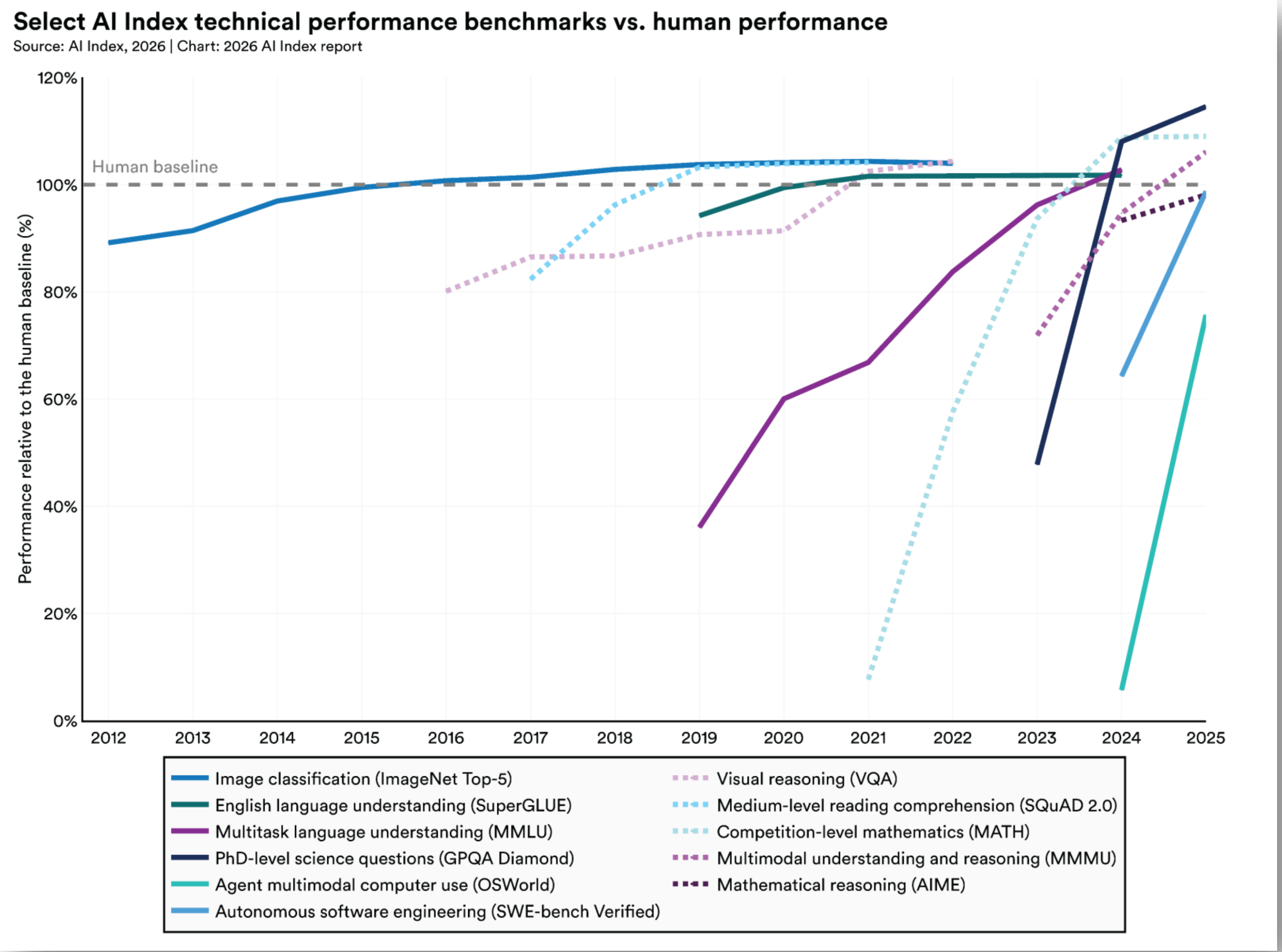
Power of AI

Matching or surpassing human performance in several intellectual benchmarks



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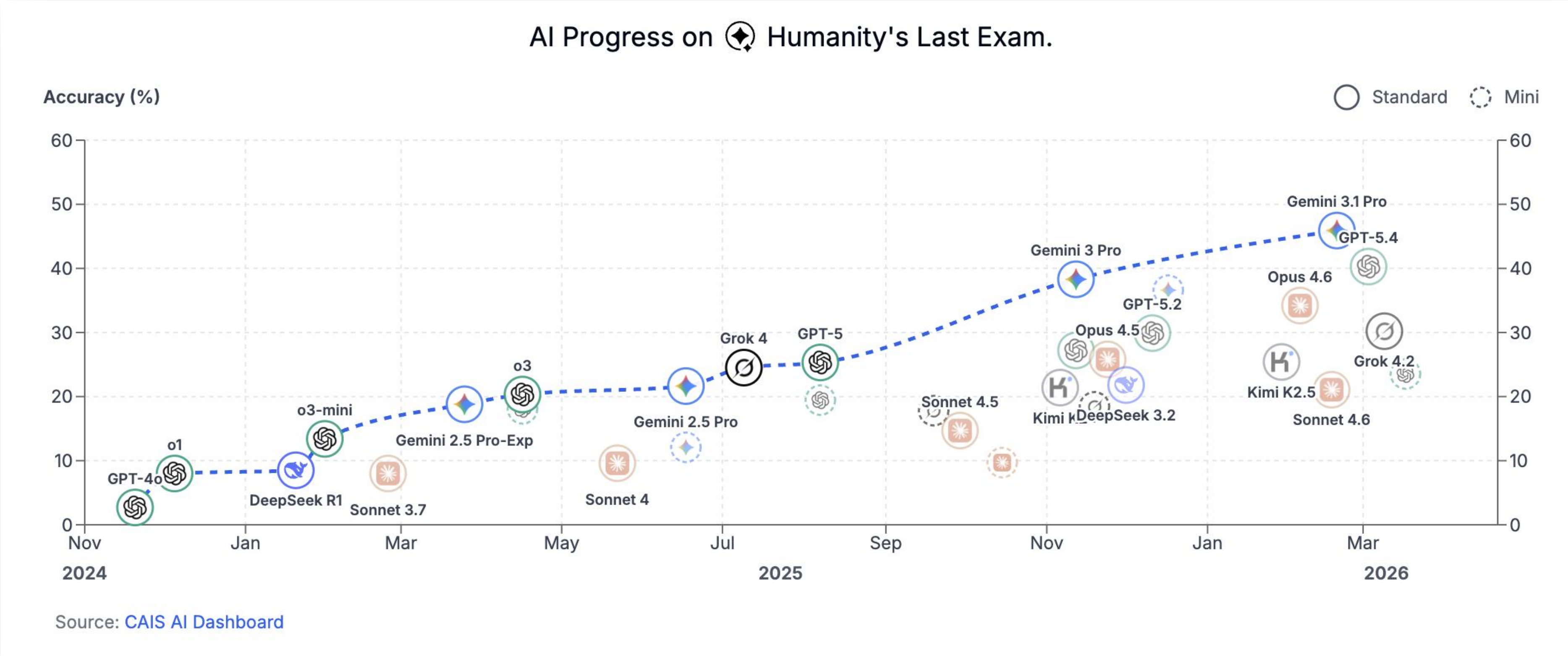
Humanity’s Last Exam



Humanity's Last Exam (agi.safe.ai)

Power of AI

Software Development

- SWE-bench Verified ~100%

- Vibe Code Bench ~57%

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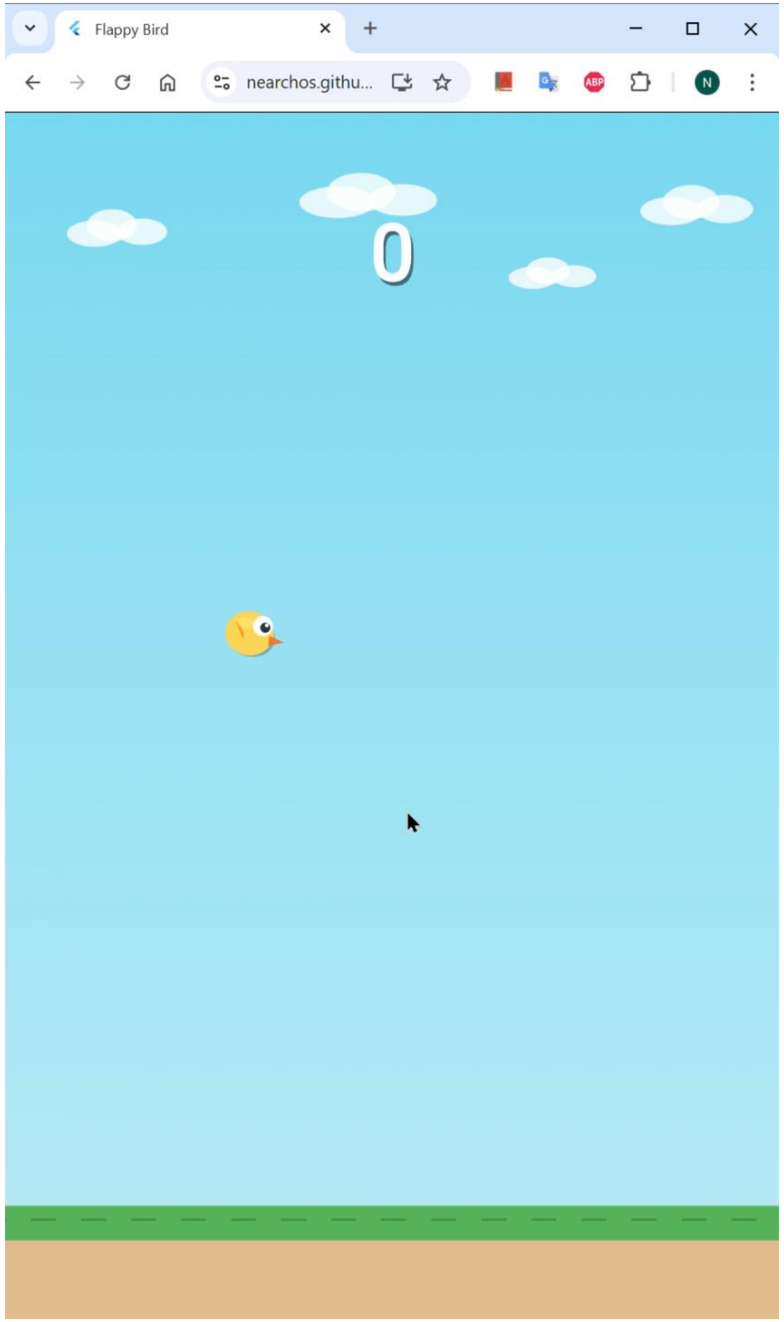
Power of AI

Software Development

- SWE-bench Verified ~100%
- Vibe Code Bench ~57%
- How about building games?



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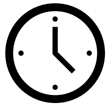
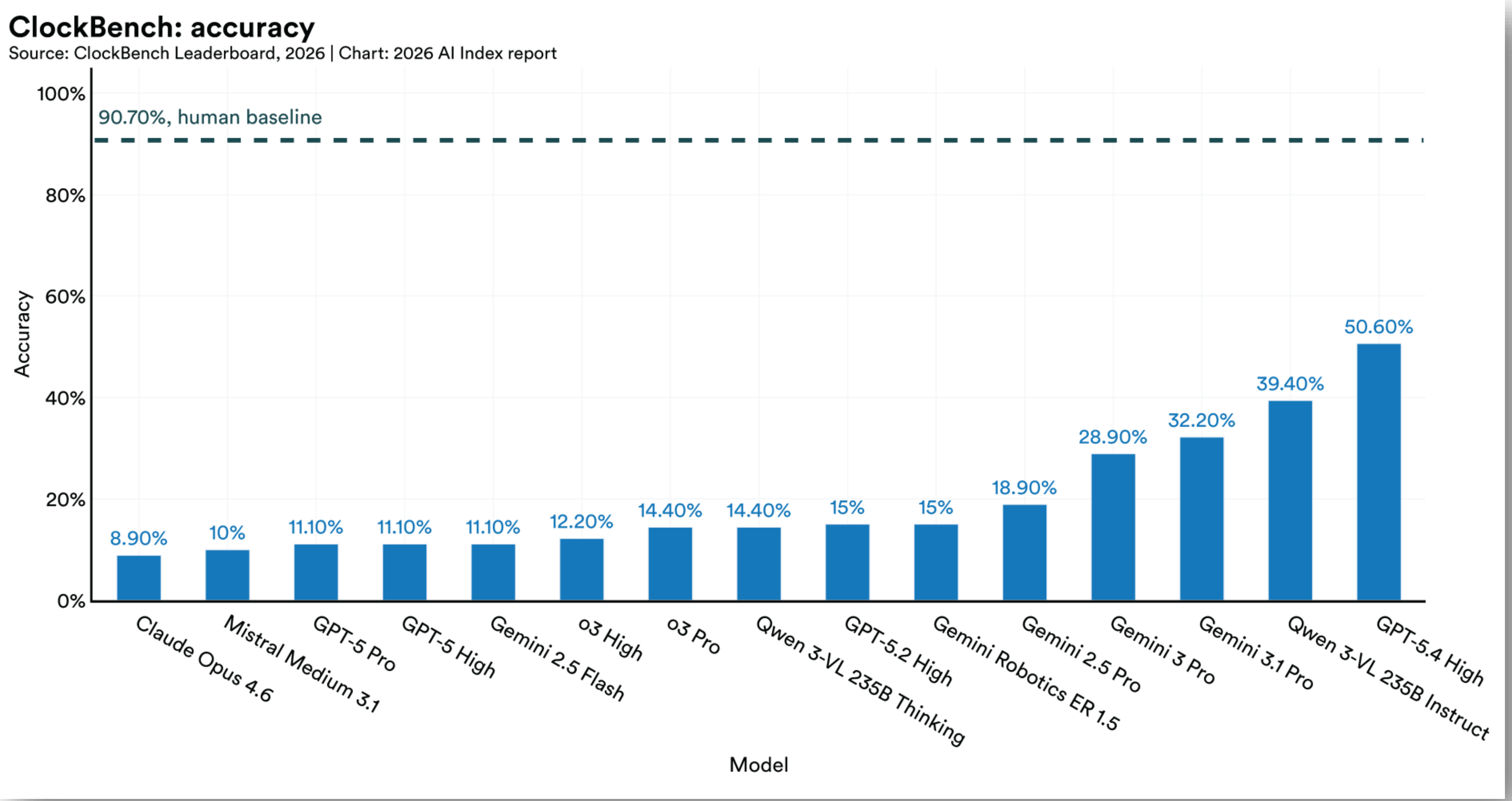
Paspallis, N. "*How well can AI chatbots develop full apps? Recreating the Flappy Bird game using a single prompt issued to LLM-based AI chatbots*" (under review)

Power of AI

Limitations!

What's the  ?

Jagged Intelligence



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Risks of AI

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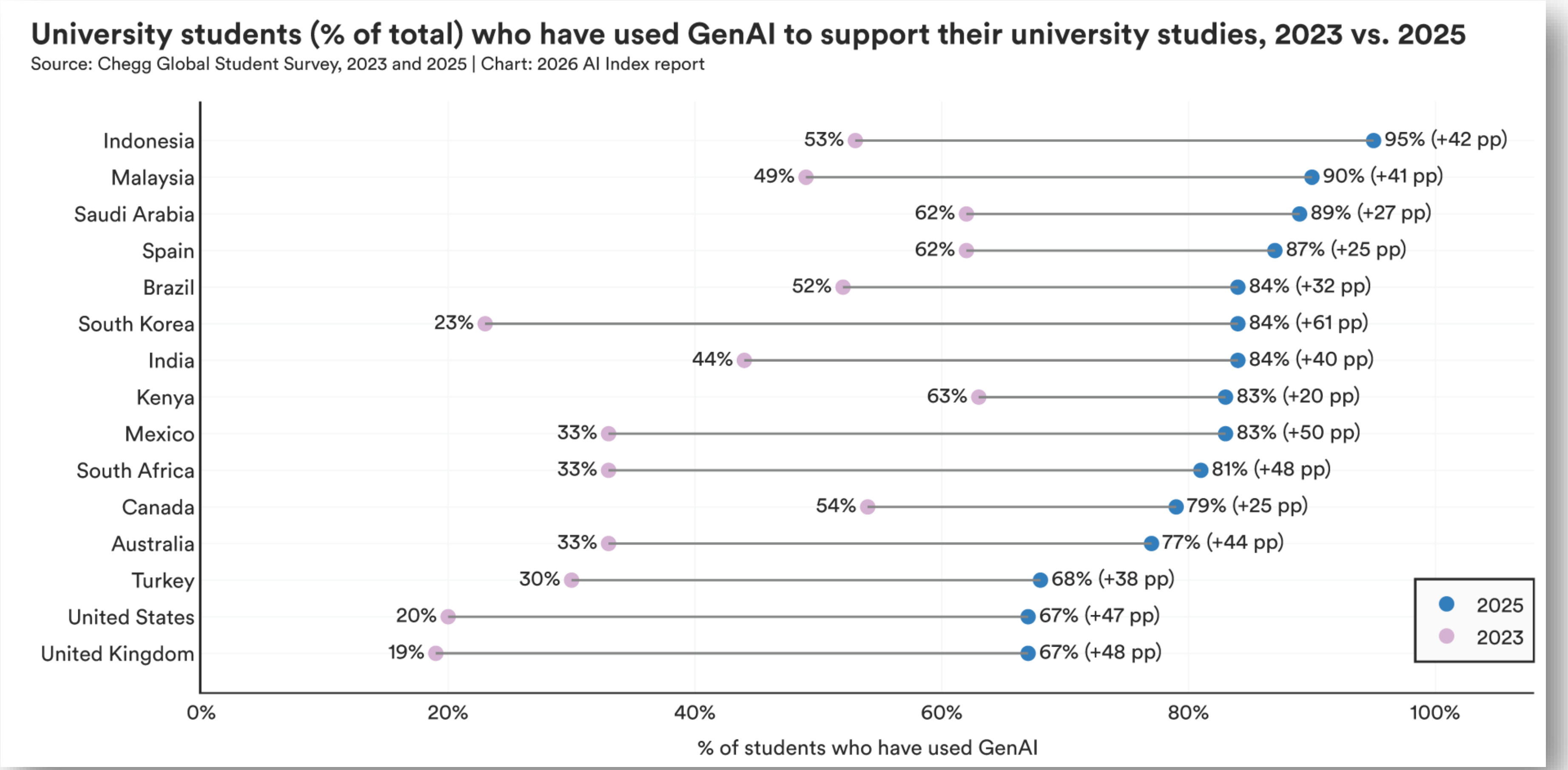
Risks of AI: Education

Overuse and misuse in education

AI increases the risk that "***learners will miss the journey!***"



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Risks of AI: Thinking atrophy

“[over-reliance on AI tools] *can lead to deterioration of cognitive faculties that ought to be preserved.*”

The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers

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Abstract

The rise of Generative AI (GenAI) in knowledge workflows raises questions about its impact on critical thinking skills and practices. We survey 319 knowledge workers to investigate 1) when and how they perceive the enaction of critical thinking when using GenAI, and 2) when and why GenAI affects their effort to do so. Participants shared 936 first-hand examples of using GenAI in work tasks. Quantitatively, when considering both task- and user-specific factors, a user's task-specific self-confidence and confidence in GenAI are predictive of whether critical thinking is enacted and the effort of doing so in GenAI-assisted tasks. Specifically, higher confidence in GenAI is associated with less critical thinking, while higher self-confidence is associated with more critical thinking. Qualitatively, GenAI shifts the nature of critical thinking toward information verification, response integration, and task stewardship. Our insights reveal new design challenges and opportunities for developing GenAI tools for knowledge work.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI.**

Keywords

Critical thinking, Generative AI tools, Knowledge worker, Bloom's taxonomy, Survey

ACM Reference Format:
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1 Introduction

Generative AI (**GenAI**) tools, defined as any *"end user tool [...] whose technical implementation includes a generative model based on deep learning"*,[1] are the latest in a long line of technologies that raise questions about their impact on the quality of human thought, a line that includes writing (objected to by Socrates), printing (objected to by Trithemius), calculators (objected to by teachers of arithmetic), and the Internet.

Such consternation is not unfounded. Used improperly, technologies can and do result in the deterioration of cognitive faculties that ought to be preserved. As Bainbridge [7] noted, a key irony of automation is that by mechanising routine tasks and leaving exception-handling to the human user, you deprive the user of the routine opportunities to practice their judgement and strengthen their cognitive musculature, leaving them atrophied and unprepared when the exceptions do arise.

In response, research has begun looking closely at how different activities are impacted by GenAI and the extent to which cognitive offloading [8] occurs, and whether this may be an undesirable thing. Some work has focused, for instance, on studying the effects of GenAI use on memory (e.g., [1, 106]) and on creativity (e.g., [28, 100]). Moreover, design research has also been developing interventions that *improve* the ability of people to think in certain ways (e.g., [24]). We review these lines of work in Section 2.

In this paper, we focus on a higher-level concept that captures another aspect of thought considered desirable and worthy of preservation: *critical thinking* (defined in Section 2). The effect of the use

[1]While there is no broad consensus on how to define this now-common term, for clarity we adopt this definition, a rationale for which is given in [115].

ACM ISBN 979-8-4007-1394-1/25/04
https://doi.org/10.1145/3706598.3713778

Hao-Ping (Hank) L., et al. 2025. (CHI '25)

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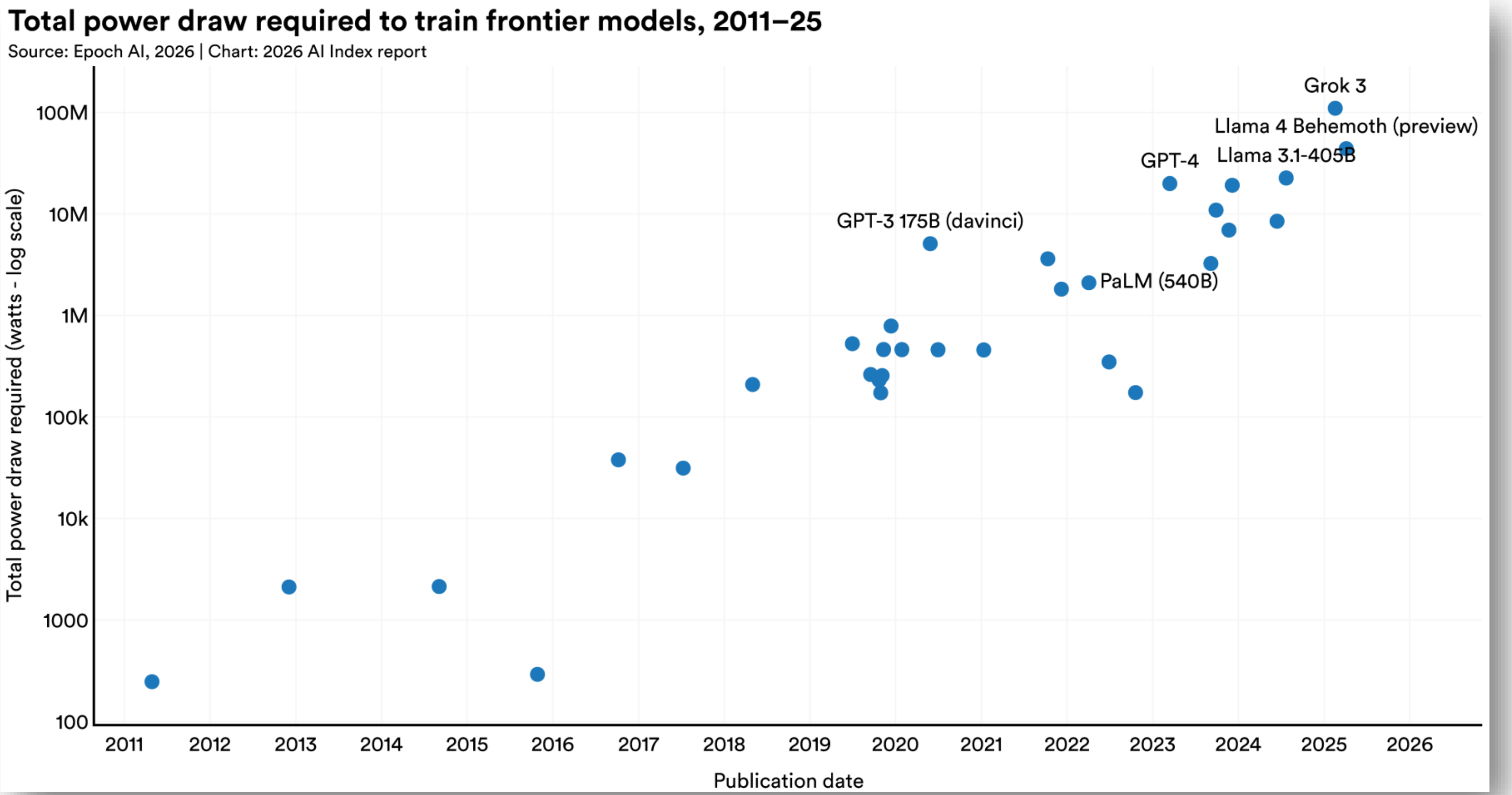
Risks of AI: Environment

Energy and Environment

Training Grok 3 equals 9 days of CY power



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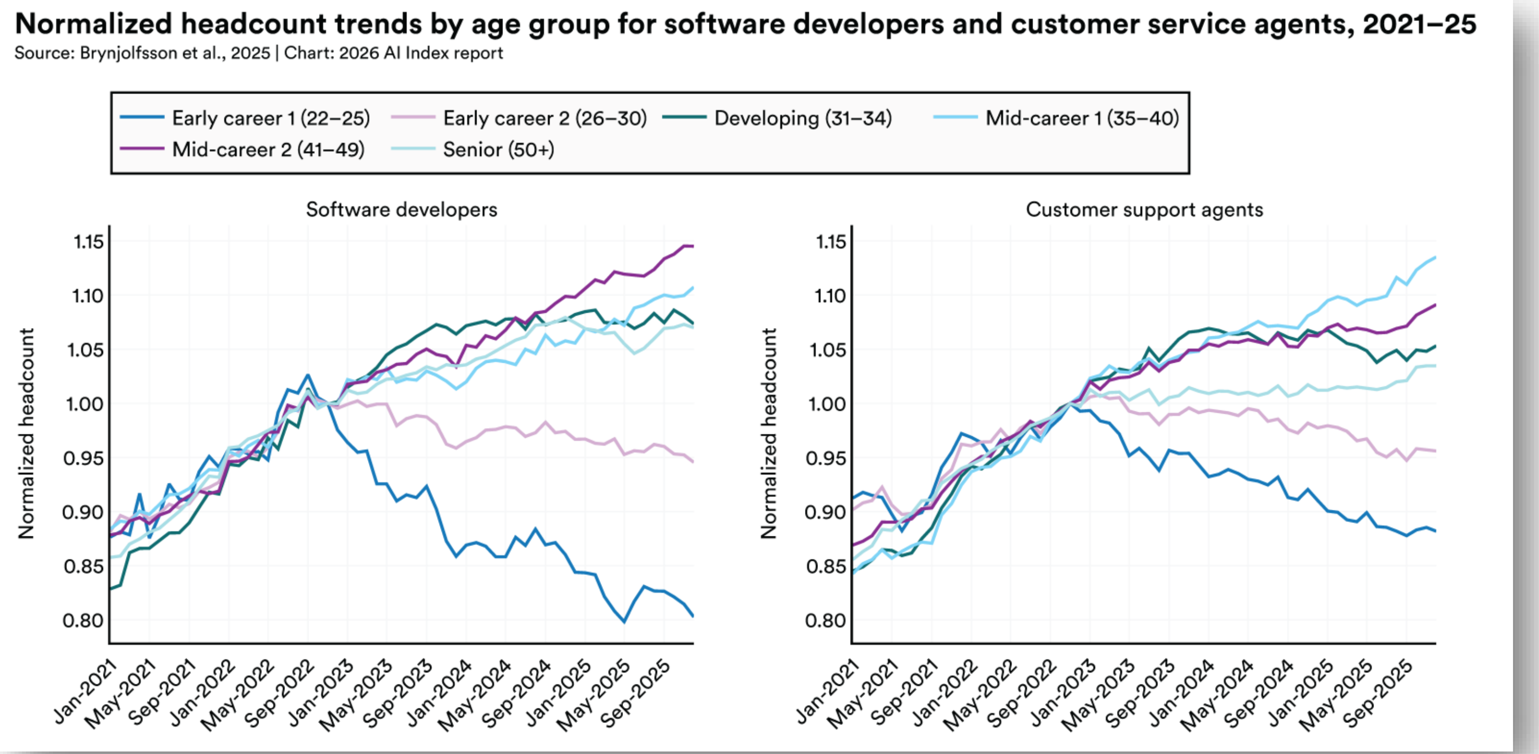
Risks of AI: Employment

Wealth concentration

Universal basic income



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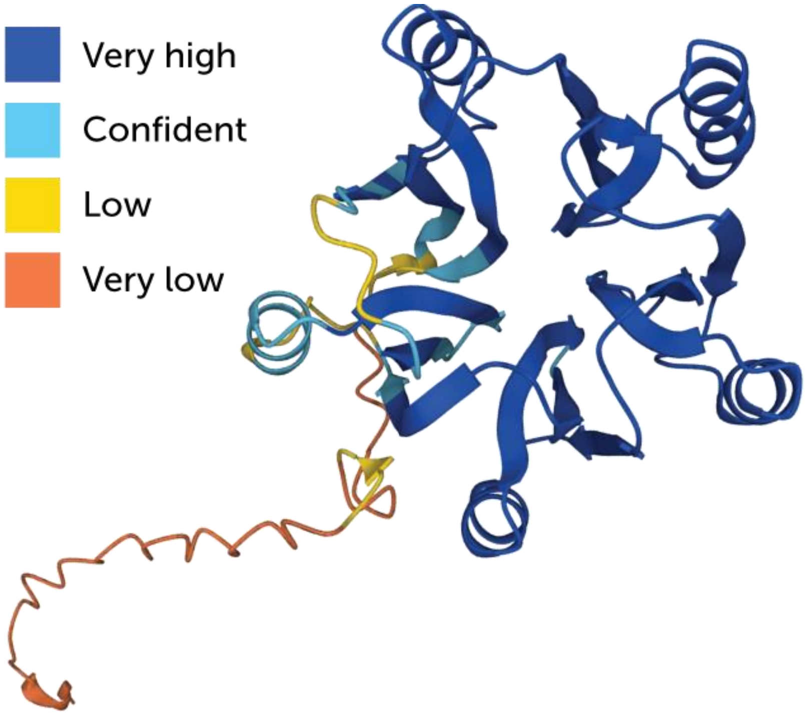
Should we be concerned?

Vigilant and optimistic

AlphaFold



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Thank you!